

ETS052

Lecture 6

2006

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Ethernet and IEEE 802.3

MAC header field type/length

- ◆ If < 600 (hex) then 802.3 frame
- ◆ If ≥ 600 (hex) then Ethernet frame
- ◆ Border defined by XEROX

Ethernet Dates

1973

- ◆ Bob Metcalfe invents Ethernet

1976

- ◆ Ethernet standard set by Digital, Intel and XEROX
- ◆ DIX Ethernet

OFDM

Orthogonal Frequency-Division Multiplexing

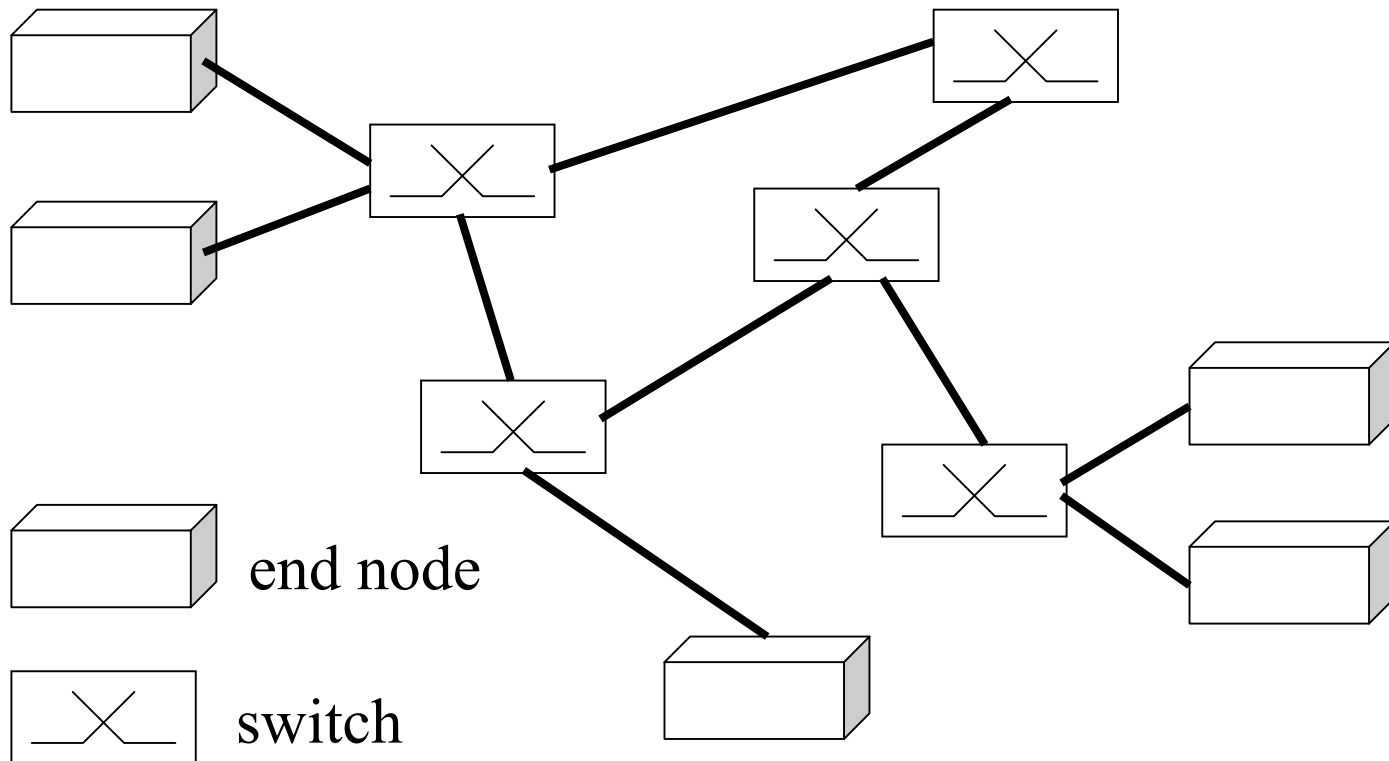
- ◆ FDM
- ◆ One sender use all sub bands
- ◆ Given n sub bands, n bits can be sent simultaneously

Lecture 6 Contents

- ◆ Switching
- ◆ WAN
- ◆ Net of LANs
- ◆ Network Layer
- ◆ Transport Layer
- ◆ Connecting devices
- ◆ Layered Models
- ◆ Fragmentation

Why Switching

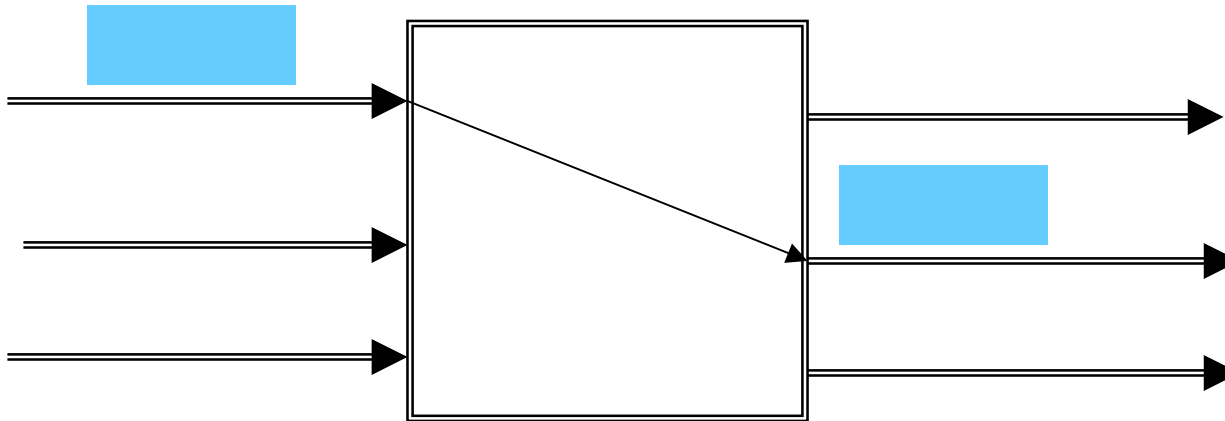
Connecting LANs



Switch

Frames arrive; switch forwards frame towards destination

”Connects” incoming link with outgoing link



Switching Networks

- ◆ Circuit switched
 - Allocated full path for whole session
- ◆ Packet switched
 - Split data into packets (cells/datagram/frames)
 - Datagram approach
 - Virtual Circuit
- ◆ (Message switched)

WAN backbones

Connecting LANs

- ◆ Compare topologies with LAN topologies
- ◆ Point to point
- ◆ Bus
- ◆ Star
- ◆ Ring

Problem Connecting LANs



Problem Connecting LANs

All connected LANs must have

- ◆ same frame format
- ◆ same address structure
- ◆ unique addresses for ALL end nodes
- ◆ switches that know of all used addresses and their locality

Solve the addressing and framing problems

Add a network layer:

- ◆ Network IDs
- ◆ End node IDs
- ◆ Datagram encapsulated in link frames

Duties

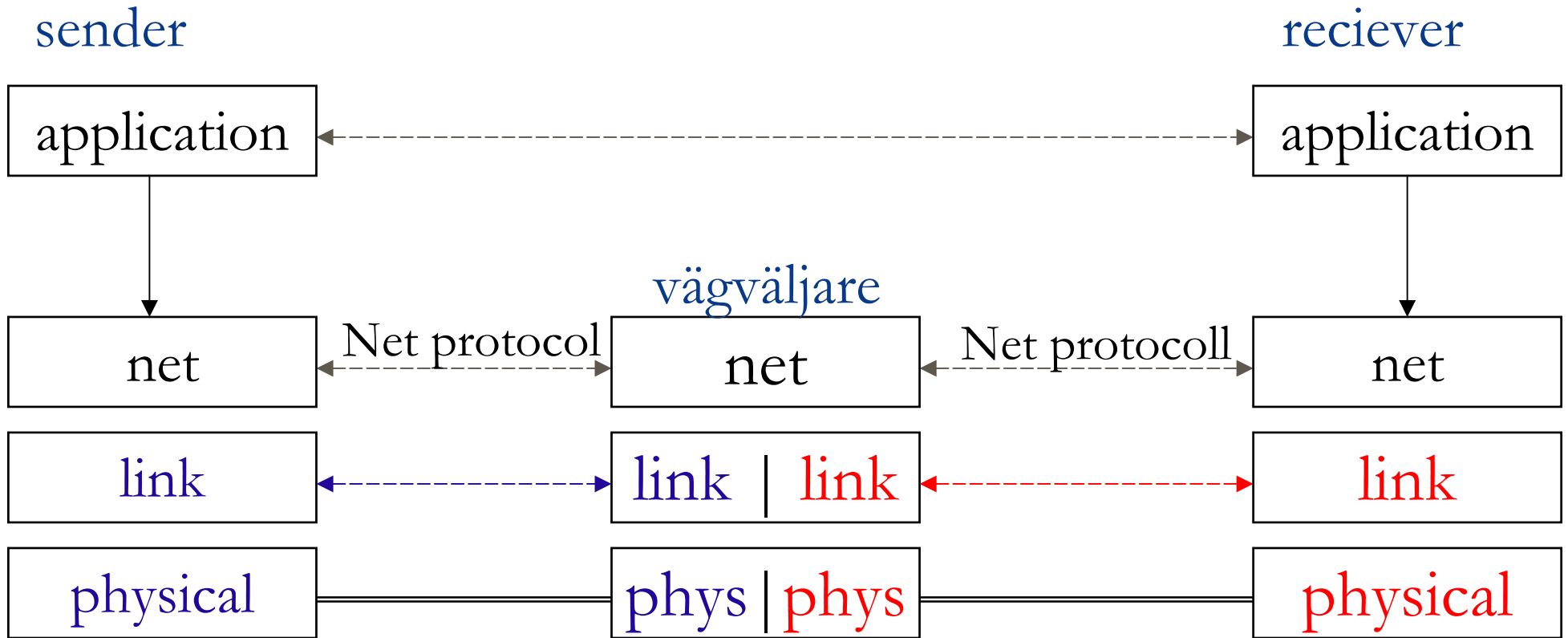
- ◆ Path selection
- ◆ Addressing

New problem!

- ◆ Mapping addresses on network layer to link layer addresses
- ◆ Solved in Lecture 7. Stay Tuned!

Best Effort Delivery

Network Layer



More layers?

Network layer

- ◆ Global communication node to node

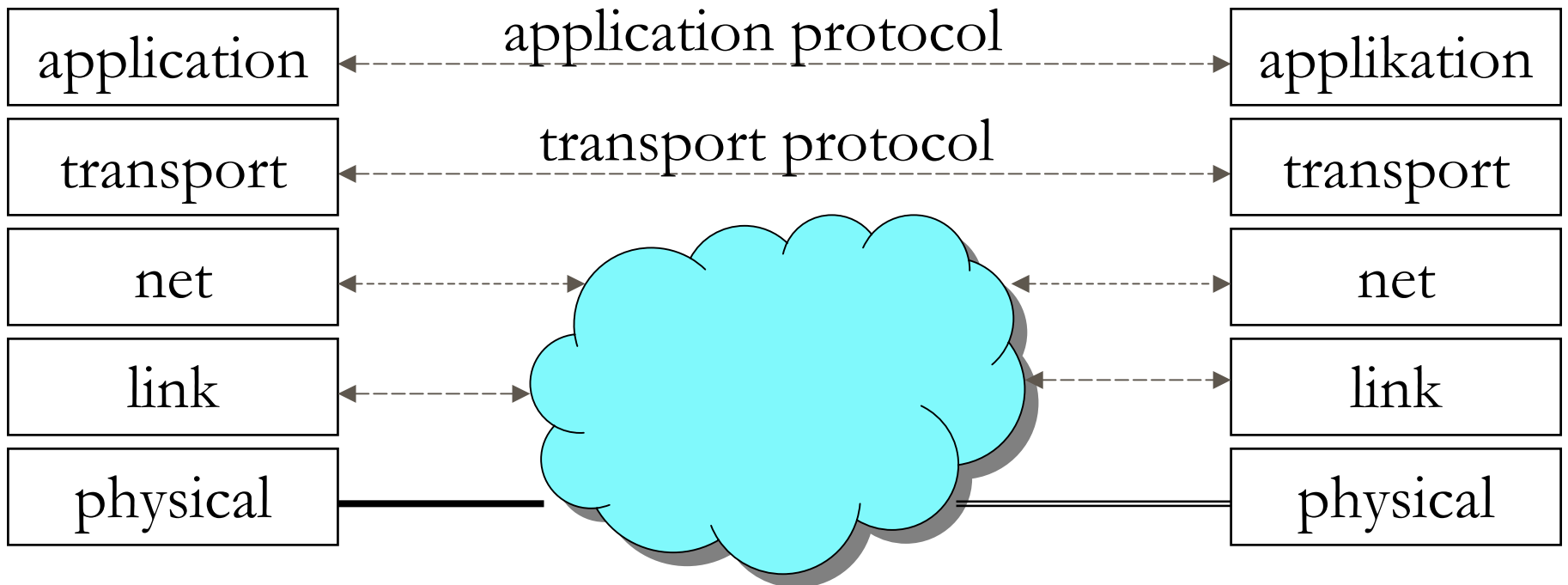
Need for process to process communication

- ◆ A node must be able to run several processes/applications

Enter Transport Layer

Transport Layer

- ◆ Process to process delivery
- ◆ Independent of network structure
- ◆ Process IDs



Transport Layer

Connection oriented

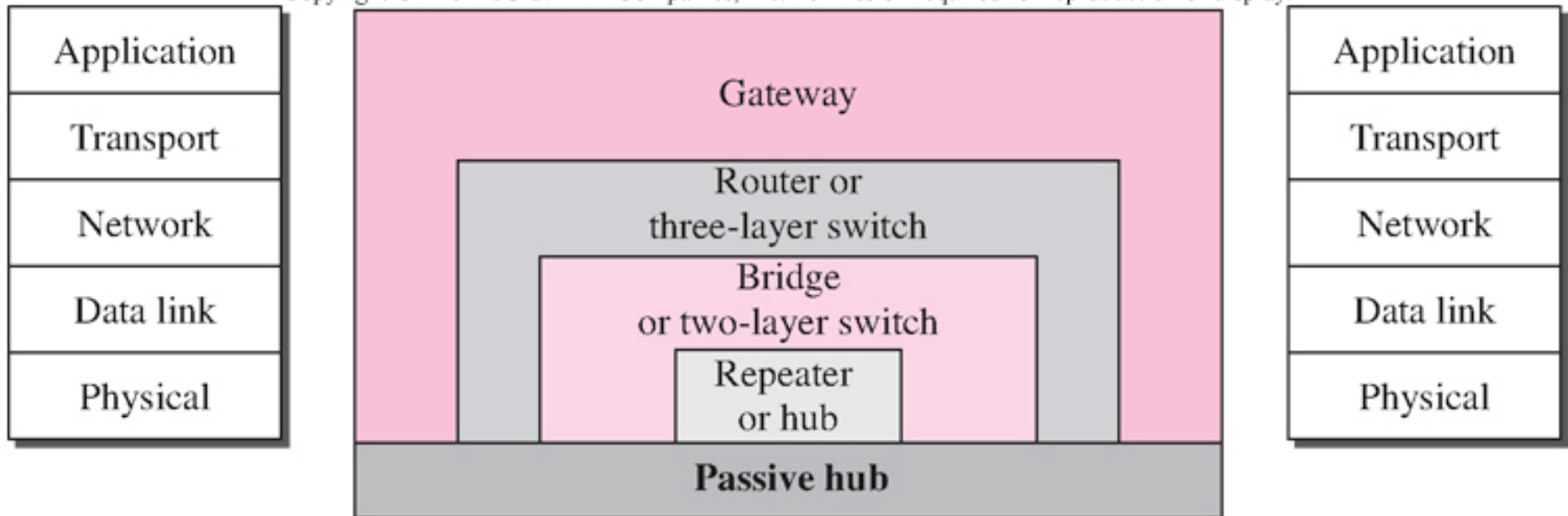
- ◆ Setup – Tear Down of session
- ◆ Error correction
 - Goal: data delivery in correct order and with no errors
 - Retransmission of packets with errors
- ◆ Flow control

Connectionless

- ◆ Just adds multiplexing

Connecting devices

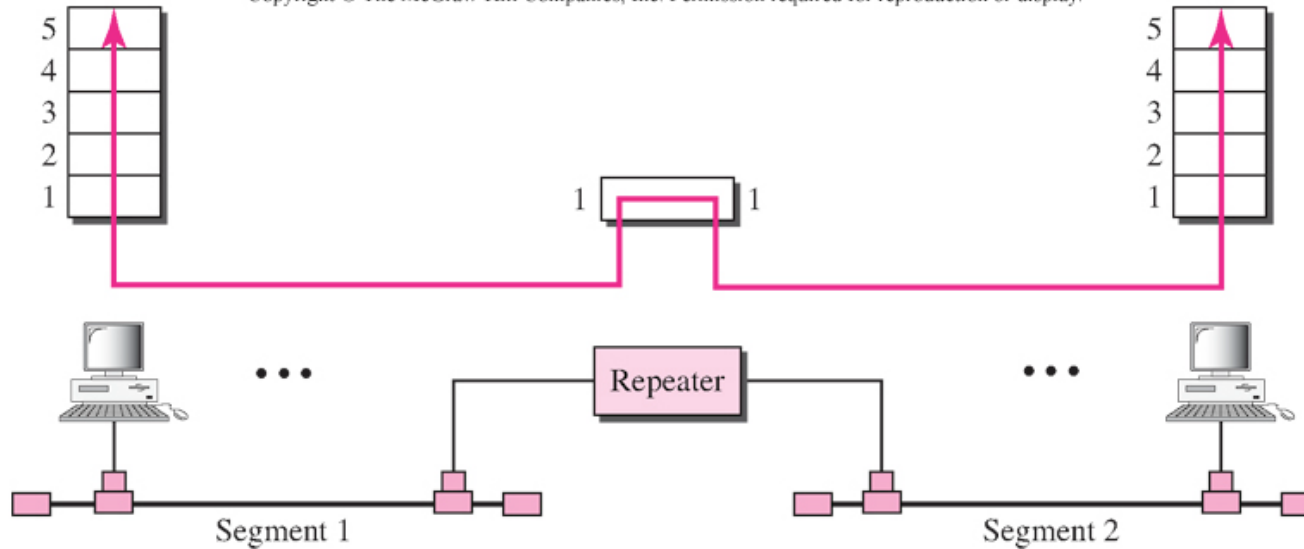
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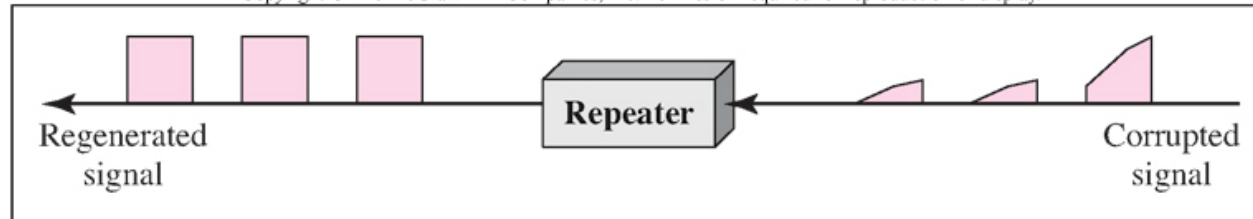
Note! "Switch" can be used on several layers

Connecting device: Repeater

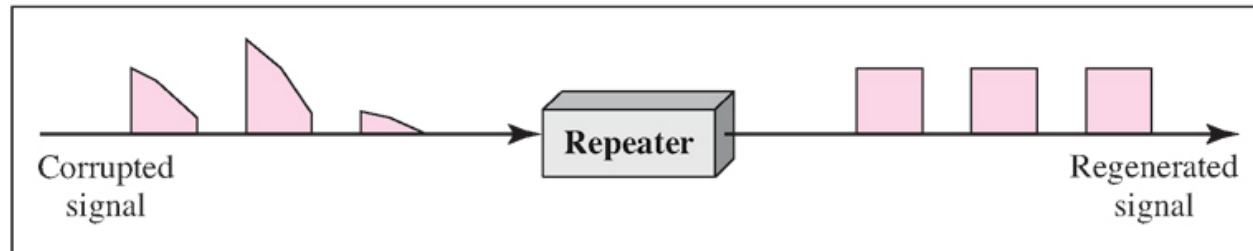
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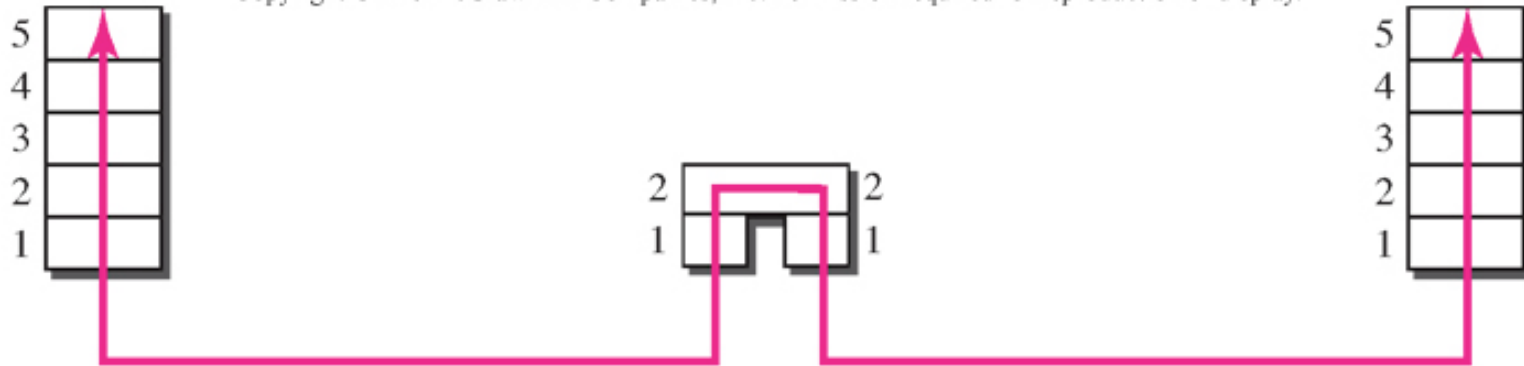
a. Right-to-left transmission.



b. Left-to-right transmission.

Connecting devices: Bridge

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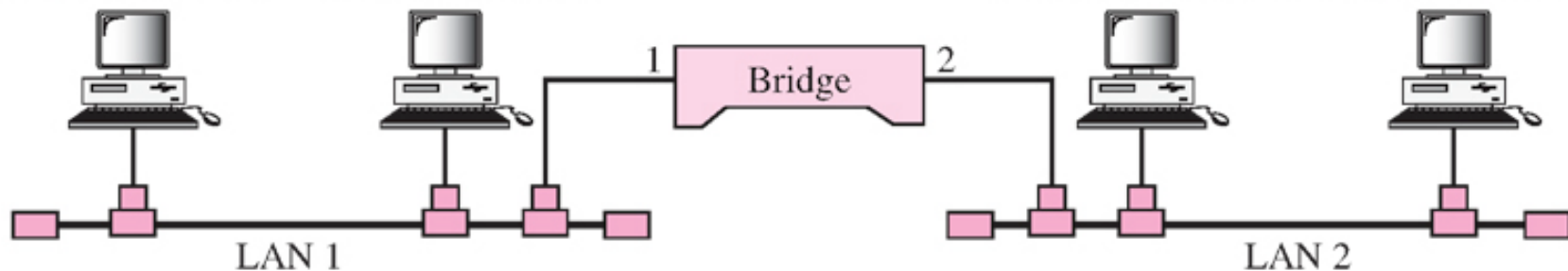


Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	1
64:2B:13:45:61:12	2
64:2B:13:45:61:13	2

Bridge Table

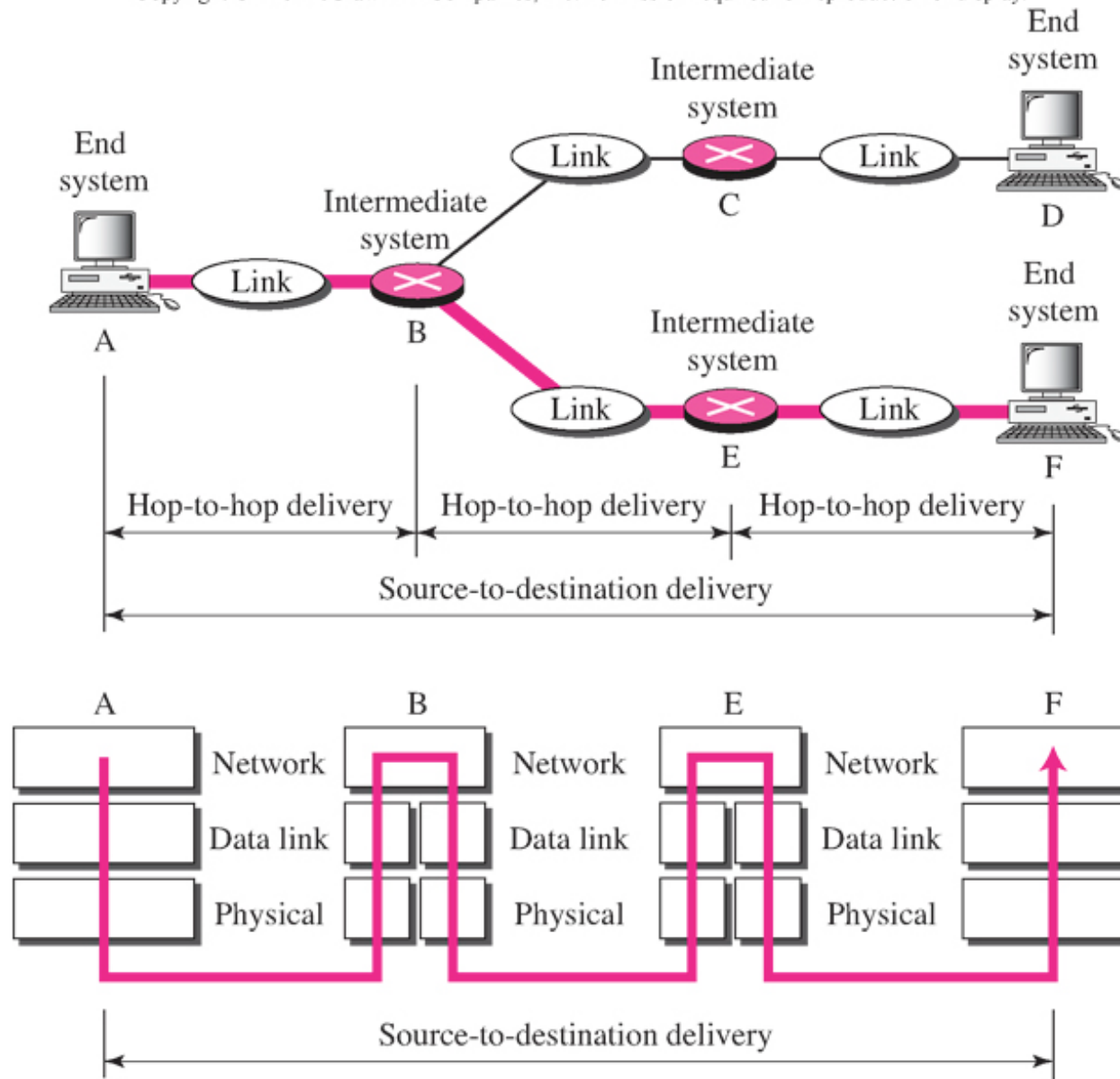
71:2B:13:45:61:41 71:2B:13:45:61:42

64:2B:13:45:61:12 64:2B:13:45:61:13



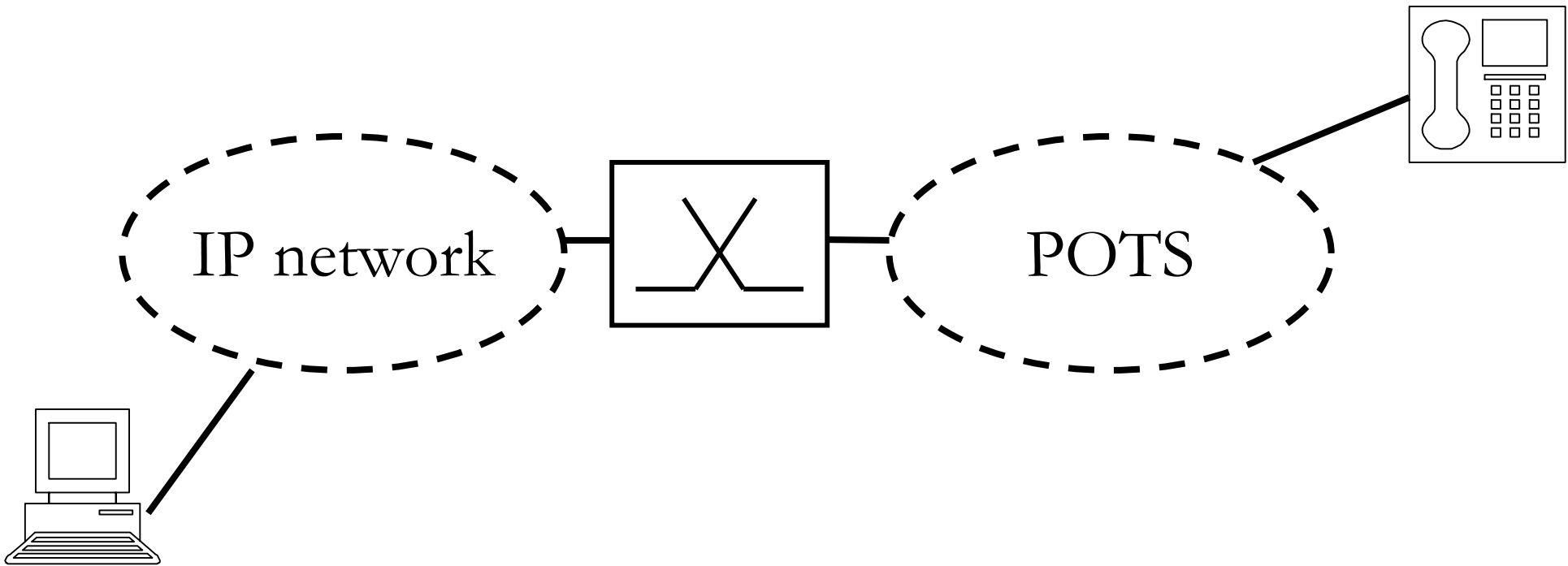
Connecting device: Router

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Connecting devices: Gateway

Interconnects applications



Layered models

Goal:

- ◆ Process to process communication over networks with different links and many intermediate nodes and links

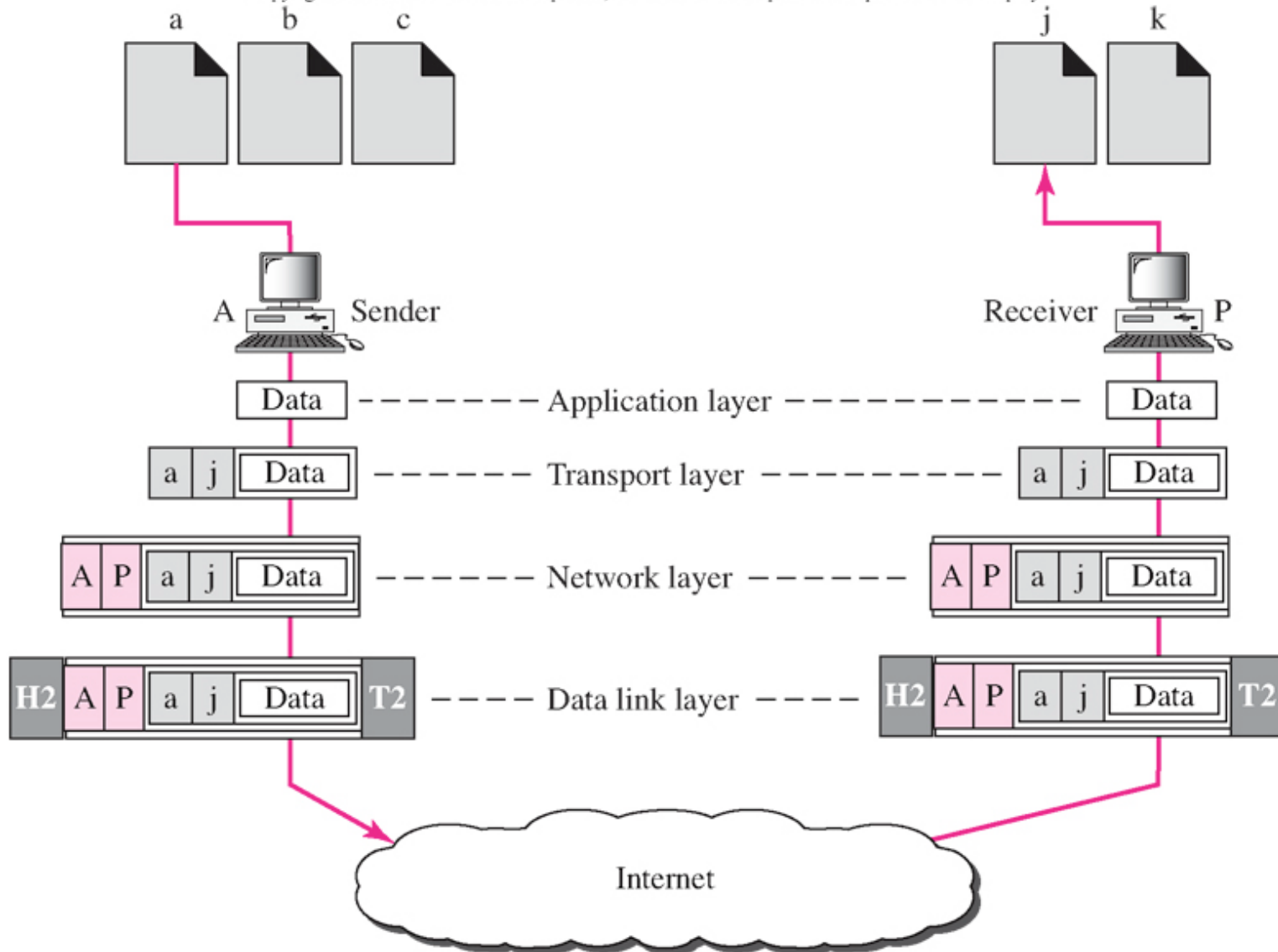
Divide task into layers

Each layer has its specific task

Layer only have to interface to layer above and below

Layered models: Encapsulation

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Fragmentation

Split data/segment/datagram/frame into smaller “packets”

Needed if data does not fit into frame, datagram, segment

Routers may fragment if max frame length on “next link” is smaller than datagram length.

End node defragments.

OSI layered model

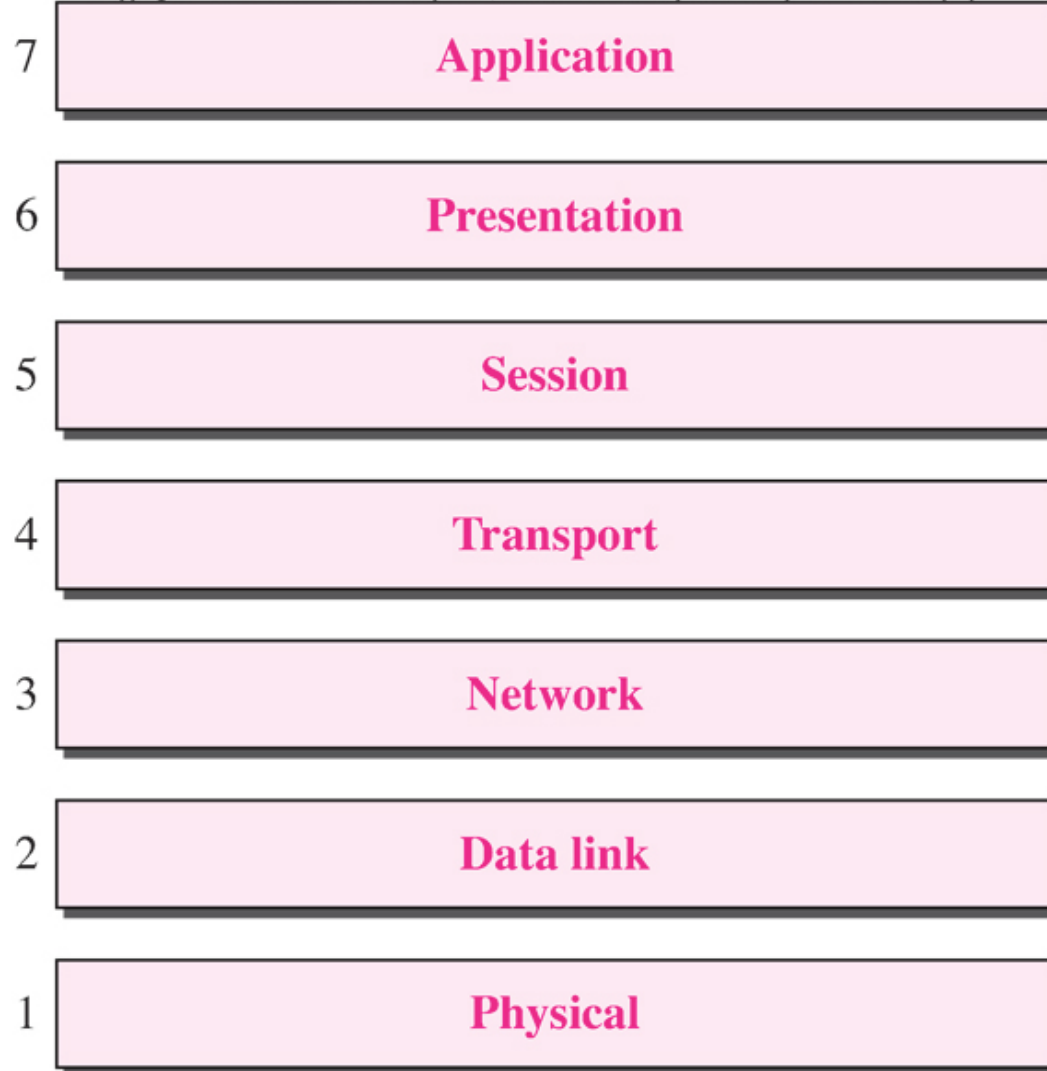
Open Systems Interconnect

Created by ISO; beginning 1970s

This is a model, not a protocol

OSI model

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Transmission Modes

Simplex

- ◆ One transmitter, one or more receivers. Ex Radio, TV

Half Duplex

- ◆ Both stations transmit and receive, but only one transmitter can talk at the same time. Ex SSB radio

Full Duplex

- ◆ Both stations can transmit and receive simultaneously

Error, flow control in transport layer?

Why?

Error, flow control in transport layer

